Mixed quadratic and cubic derivative singlets
A local reduction to the multiprobe chord law

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Abstract

The quadratic and cubic derivative singlets constructed in the preceding notes obey the same joint fixed-degree moment rule as two independent probe species of lengths two and three. The microscopic operators share the Hamiltonian couplings; the reduction follows from cancellations of marked Wick diagrams. We classify every local component into fourteen templates and give an exact polynomial certificate. This supplies the missing interface to the existing multiprobe four-point function. We evaluate the mixed out-of-time-order correlator, restore its mutual crossing factor q^6, and obtain its short-time commutator growth.

1 Joint law and physical target

Use standard CAR, the complex Gaussian number-preserving p-body Hamiltonian including overlapping index sets, and the annealed normalized trace $\tau = \mathbb{E}\,\mathrm{tr}$ at $\beta = \mu_{\mathrm{chem}} = 0$. With $\hbar = 1$, take

$$N, p \longrightarrow \infty, \quad \lambda = p^2/N > 0, \qquad q = e^{-4\lambda}, \qquad \sigma^2 = \mathcal{J}_s^2/\lambda.$$

Let $h = H/\sigma$, $X = Y/\sqrt{W}$ and $\widehat{R} = R/\sqrt{V_R}$ be the normalized Hamiltonian, quadratic singlet, and cubic residual defined on page 2. Their joint limiting moments are

$$\boxed{\tau(A_1 \cdots A_n) = \sum_{\pi \text{ pairs identical species}} \prod_{\substack{\{e,f\}\subset\pi \\ e \text{ crosses } f}} q^{d_e d_f}, \qquad (d_h, d_X, d_{\widehat{R}}) = (1,2,3).} \tag{1}$$

In particular,

$$\tau\Big(X\widehat{R}^2\Big) = 0, \quad \tau\Big(X^2\widehat{R}^2\Big) = 1, \quad \tau\Big(X\widehat{R}X\widehat{R}\Big) = q^6, \quad \tau\Big(\widehat{R}^4\Big) = 2 + q^9. \tag{2}$$

Thus distinct singlets already have a nonzero equal-time commutator: $\tau\Big([X,\widehat{R}]^\dagger[X,\widehat{R}]\Big) = 2(1-q^6)$. The established multiprobe framework [1, 2] now applies to these specific composite insertions. Our task is the reduction to that framework, not a replacement four-point kernel.

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2 Operators and local marked weights

Put $L = [H, \cdot]$, $\mathsf{J}_H B = \{H, B\}/2$ and

$$A = \sum_i (Lc_i)^\dagger Lc_i, \quad M_3 = \sum_i c_i^\dagger L^3 c_i, \quad T_3 = (M_3 + M_3^\dagger)/2.$$

The preceding construction [3] uses ordinary Gaussian coupling chaos $\mathcal{W}_j$, not energy normal ordering. At finite N, $Y_N = A - a_0 - c_N(H^2 - C)$ subtracts the scalar expectation and its quadratic Hamiltonian projection, with the coefficients defined sector by sector. Its DS limit is Y, and

$$c_N \longrightarrow \frac{2\lambda(1-q)}{1+q}, \qquad W = 4\mathcal{J}_s^4 \left[\lambda^{-1} + \frac{4q}{1+q}\right].$$

For $(T, S, Z) = \mathcal{W}_3(T_3, \mathsf{J}_H A, H^3)$ set

$$R = T - aS - bZ, \qquad a = -\frac{(1-q)^2}{1+q^2}, \tag{3}$$

$$b = \frac{2\lambda(1-q)(1+q^3)}{(1+q^2)(1+q+q^2)}, \qquad D = (1+q^2)(1+q+q^2). \tag{4}$$

Its variance is $V_R = \sigma^6 G_R / D^2$, where

$$G_X = (1+q)^2 W/\sigma^4 = 4\lambda(1+q)^2 + 16\lambda^2 q(1+q), \tag{5}$$

$$G_R = 4\lambda(1-q^2)(1+q^2)(1+q+q^2)^2 + 16\lambda^2 q^2(1+q)(1+q^2)(1+q+q^2)(2q^2-q+2). \tag{6}$$

The symbol `X` in the raw polynomial code denotes Y, with the denominator $1+q$ cleared; `R` has denominator D cleared. Every underlying Hamiltonian slot contributes σ.

Expand a composite into consecutive slots. A quadratic vertex has one possible mark, 12; a cubic vertex has marks $12, 23, 13$. After clearing denominators their weights are

vertex	mark coefficients	unmarked coefficient
Y	$1+q$	$-2\lambda(1-q)$
R	$-q(1+q+q^2),\ -q(1+q+q^2),\ D$	$-2\lambda(1-q)(1+q^3)$

There is at most one mark per vertex. Internal Gaussian pairings within each composite block are deleted before the DS limit. The exact identity $A = (2p - \mathfrak{D})H^2$, $\mathfrak{D}B = QB - \sum_i c_i^\dagger B c_i$, and its three-slot counterpart produce these marks, including the nonadjacent 13 choice.

3 Why fourteen local templates suffice

For an underlying pairing π, let $c(\pi)$ count ordinary chord crossings. The previously established fixed-word determinant expansion [3] gives, for selected marks $j = (a, b)$,

$$q^{c(\pi)} \exp\left[\lambda \sum_j \alpha_j z_j + \lambda \sum_{j<k} \delta_{jk} z_j z_k\right], \qquad \alpha_j = 2 - 4\chi_j. \tag{7}$$

Here χ_j tests whether the two marked chords cross. For $j = (a, b)$ and $k = (c, d)$, $\delta_{jk} = 4$ precisely when $\pi(a) = d, \pi(b) = c$; otherwise it is zero. Extract each selected mark once and insert the coefficients on page 2. The exponential has no higher connected mark term: triple set overlaps vanish in the fixed-word DS limit. Fixed-order overlap exponential bounds justify the derivatives.

Component classification. Make a graph whose vertices are the composite blocks, joining two blocks when at least two Hamiltonian chords connect them. A vertex has two or three slots, so it has at most one such neighbor. Every component is a singleton or a pair. Ordinary one-slot H vertices remain spectators. Fix the component's external endpoints and sum over all assignments to its internal slots. All crossings with chords having no endpoint in the component factor out: the component slots are consecutive blocks.

An outside mark cannot change under these assignments. If both of its chords ended in one component vertex, it would itself be a double-bridge neighbor, already in that component. If the partners end in distinct vertices, their cyclic block order is fixed. A delta term connecting the component to an outside mark would likewise require a double bridge. Thus all outside marked factors are constant during the local sum. Crossings among the component's own incident chords are retained in the local template, including its residual external endpoints.

local component	templates	local pairings	sum
singleton Y	1	2	0
singleton R	1	6	0
YY, two bridges	1	2	G_X
YR or RY, two bridges	4	24	0
RR, two bridges	6	108	0
RR, three bridges	1	6	G_R

For two blocks, external endpoints can lie on either separating arc. The six partial RR templates are two endpoint color orders times three arc splits. These possibilities exhaust the topology. Exact determinant polynomials give the displayed sums for independent symbols q, λ. Every incomplete component cancels. The survivors pair whole equal-species blocks. Compressing such pairs gives $d_e d_f$ elementary crossings, and dividing by G_X, G_R proves (1) for every fixed word.

4 Real-time mixed interaction observable

Define $X(t) = e^{itH} X e^{-itH}$ and

$$F_{23}(t) = \tau\Big(X(t)\widehat{R}X(t)\widehat{R}\Big), \qquad D_{23}(t) = \tau\Big([X(t),\widehat{R}]^\dagger[X(t),\widehat{R}]\Big). \tag{8}$$

The uncrossed mixed pair is $\tau\Big(X(t)^2\widehat{R}^2\Big) = 1$. Trace cyclicity and Hermiticity therefore give $D_{23}(t) = 2[1 - \mathrm{Re}\, F_{23}(t)]$. For distinct species there is only the single allowed crossed pairing in F_{23}, rather than the three channels of a same-species correlator.

Here is a positive-series form of BINT Eq. (4.10) [1], in symmetric Jacobi normalization. Write $a = q^2$, $b = q^3$, $Q_{23} = q^6$ and

$$J_{n-1,n} = \sigma\sqrt{\frac{1-q^n}{1-q}}, \qquad U(t) = e^{-itJ}, \qquad v_n = U_{n0}(t), \tag{9}$$

$$c_i = \frac{(b^2;q)_i}{(q;q)_i}a^i, \qquad V_{li} = \sqrt{c_i}\, b^l \sqrt{\frac{(q;q)_{l+i}}{(q;q)_l}}\, v_{l+i}. \tag{10}$$

With $(x;q)_i = \prod_{j=0}^{i-1}(1-xq^j)$,

$$\boxed{F_{23}(t) = Q_{23}\sum_{n,i\geq 0} a^n |(U(t)^\dagger V)_{ni}|^2.} \tag{11}$$

BINT explicitly suppresses the overall mutual-probe factor in that section. Restoring Q_{23} is essential: $F_{23}(0) = q^6$, not one. BNR Section 8.1 [2] supplies the neutral complex-model interface. In (11), the four alternating real-time propagators have combined into a modulus square; this is not an extra approximation.

Direct compressed chord enumeration gives

$$F_{23}(t) = q^6 - 2q^6(1-q^2)(1-q^3)(\sigma t)^2 + c_4(q)(\sigma t)^4 + O(t^6), \tag{12}$$

$$c_4(q) = \frac{q^6(q-1)^3(q+1)(q^2+q+1)}{6} \times (9q^5 + 12q^4 + 9q^3 - 3q^2 - 12q - 11), \tag{13}$$

$$D_{23}(t) = 2(1-q^6) + 4\sigma^2 q^6(1-q^2)(1-q^3)t^2 + O(t^4). \tag{14}$$

In particular $D_{23}(0) = 48\lambda + O(\lambda^2)$, and the coefficient of t^2 is $384\lambda\mathcal{J}_s^2 + O(\lambda^2)$. The noncommutativity and its initial growth both survive at the first deformation order. A species-preserving Wick law does not make these ordinary commuting Gaussian variables.

5 Verification and numerical results

The initial mixed enumeration covers twelve words, 15,392 pairings and 433,964 local-minor checks, including $X\widehat{R}^2$, $X\widehat{R}X\widehat{R}$, $X^2\widehat{R}^2$, $\widehat{R}^4$, and words with two external Hamiltonians. All differences from (1) are exact zero polynomials. The separate fourteen-template certificate uses the older determinant implementation, not the new geometric weight evaluator. It shares the pairing generator. The factorization argument on page 3 is what extends these finite checks to arbitrary fixed degree.

An independent finite-CAR computation tests all choices of local mark or no mark on 64 mixed words at $(N, p) = (3, 1), (4, 2)$. All 6,656 normalized matrix traces equal the finite determinant coefficients exactly. Among them 1,031 are nonzero, including two negative values. Coherent repeated index sets are included to avoid an all-zero sparse test. These checks validate the mark algebra, not finite-size convergence rates.

The real-time code uses $\mathcal{J}_s = 1$. It independently sums all four-arc Hamiltonian insertions through order t^6 and compares with (11) at $t = .01$, swaps the two probe labels, and enlarges each Jacobi cutoff by 16. Results at $t = 1$ are

λ	cutoff	$F_{23}(1)$	$D_{23}(1)$
.20	52	$4.26855521\,10^{-6}$	1.99999146
.10	90	$5.71877829\,10^{-5}$	1.99988562
.05	167	$5.68948591\,10^{-4}$	1.99886210

These are finite-λ values, not estimates of a first-order small-λ coefficient. Across all fifteen points, the largest cutoff and species-swap changes are $6.39\,10^{-16}$ and $3.34\,10^{-16}$, respectively. The small-time Taylor comparison differs by at most $1.27\,10^{-15}$.

Analytic truncation control. For $b = q^{d_2}$, put

$$C_b = \prod_{j=1}^{2d_2-1} (1 - q^j)^{-1}, \quad K = C_b/(1 - a), \quad R_H = 2\sigma/\sqrt{1 - q}.$$

Then $c_i \le C_b a^i$, the map $v \mapsto V$ has norm at most $\sqrt{K}$, and its rows $l \ge M$ have norm at most $\sqrt{K}b^M$. A cutoff of size L agrees with powers of the infinite Jacobi matrix up to the first path reaching the boundary. Define $T_k(z) = \sum_{j\ge k} z^j/j!$ and

$$\delta = 2T_L(R_H|t|) + 2T_{L-M}(R_H|t|) + 2b^M.$$

Splitting low and high source rows, unitarity gives $|F_{23} - F_{23}^{(L)}| \le 2Q_{23}K\delta$. We choose the larger cutoff needed for the two species orders; the reported analytic bounds are below 10^{-11} for $|\mathcal{J}_s t| \le 1$. This bounds truncation in exact arithmetic. Floating-point accuracy is assessed by the separate comparisons, not certified by interval arithmetic.

A Fixed times, limits, and the next calculation

Equation (1) is a fixed-degree DS statement. It also determines fixed real-time correlators. Here is a moment-based justification that does not assume a uniform finite-N operator norm. For a finite real-time word, approximate each unitary by its degree-k Taylor polynomial. The scalar spectral theorem bounds its $L^r(\tau_N)$ remainder by $|t|^{k+1}\|H^{k+1}\|_r/(k+1)!$ for even integer r. Noncommutative Hölder bounds the rest of the word by fixed moments of the probes. For each fixed k, all these moments converge by (1). The limiting Hamiltonian has the compact q-Hermite support $[-R_H, R_H]$, so the limiting remainder is bounded by a constant times $(|t|R_H)^{k+1}/(k+1)!$ (and fixed polynomial factors from the other Taylor approximants). It vanishes as $k \to \infty$. Thus DS is taken first, followed by polynomial degree. No growing-time or uniform $\lambda \to 0$ estimate is used here.

The local determinant limit uses fixed numbers of slots and marks. At $\mu_{\text{chem}} = 0$, the central charge window and its exponentially small complement are handled as in the preceding notes. The two-body overlap exponent and its mark derivatives are uniform in that window. Gaussian chaos subtraction removes internal pairings before the limit. The remainder coefficients in Y_N converge in the fixed moment bounds; replacing them by their limiting values therefore preserves the theorem.

Next constructive step. The saved exact coefficients through t^6 give, coefficient by coefficient,

$$\frac{D_{23}(t)}{\lambda} : \quad 48 + 384(\mathcal{J}_s t)^2 + 512(\mathcal{J}_s t)^4 + \frac{4096}{15}(\mathcal{J}_s t)^6 + \cdots \qquad (\lambda \to 0).$$

These are the first four terms of $48\cosh(4\mathcal{J}_s t)$. This suggests a concrete next target: use the existing unequal-probe first-order four-point expansion to determine the entire fixed-time coefficient, then match it to the two-species Rindler construction. Finite Taylor agreement alone is not the derivation of that function.

The present theorem covers $H, X, \widehat{R}$. It does not yet include arbitrary high-point insertions of the linear rank-filter residual F. The time-dependent mixed commutator is microscopic interaction data; selecting a bulk mass, local vertex, or spatial dictionary remains a separate physical construction. All statements here use the ensemble-averaged infinite-temperature state and the specified order of limits.

References

[1] M. Berkooz, M. Isachenkov, V. Narovlansky and G. Torrents, *Towards a full solution of the large N double-scaled SYK model*, arXiv:1811.02584v2, Sections 4.1–4.2, especially Eq. (4.10) and footnote 8.

[2] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev-Ye-Kitaev model in the double scaling limit*, arXiv:2006.13983v2, Section 8.1, Eqs. (8.1)–(8.7).

[3] A. Day and Chengheng, local project notes: *Singlet multimark*, *Cubic singlet*, *Cubic kernel*, and *Filtered spectrum* (November 2026). Exact scripts and certificates: `results/susskind-mixed-cubic`.